



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
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Skill Progress Analytics with Personalized Learning Path Recommendation Engine

Course Name :Java Programming

Course Code :CSA0925

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Problem Background

Application Domain

- E-Learning and Educational Technology
- Personalized Learning Systems
- Learning Analytics

Current Scenario

- Most learning platforms provide the same content to every learner.
- They do not analyze individual learning progress.
- Students find it difficult to identify weak areas.

Need for the Project

- To provide personalized learning recommendations.
- To improve learning efficiency.
- To track learner progress continuously.

Target Users

- Students
- Online Learners

Problem Statement

Problem Definition

- Traditional learning systems offer uniform learning content without considering learners' skill levels and progress.

Evidence / Statistics

- More than **70%** of learners prefer personalized learning experiences.
- Adaptive learning can improve learner engagement by **30–40%**.
- Personalized recommendations improve course completion rates.

Limitations of Current Systems

- Same learning content for all users.
- No personalized recommendations.
- Poor progress tracking.
- Limited learning analytics.

Expected Impact

- Better learning outcomes.
- Improved learner engagement.
- Efficient skill development.

Objectives

Primary Objective

Develop a Java-based system that analyzes learner progress and recommends personalized learning paths.

Measurable Objectives

- Track learner performance.
- Analyze strengths and weaknesses.
- Recommend personalized learning paths.
- Generate progress reports.
- Improve learning efficiency.

Project Scope

- Student Registration
- Course Tracking
- Skill Analytics
- Learning Recommendations
- Progress Dashboard

Literature survey

S.No	Author Names	Year	Research Problem	Tools / Techniques Used	Research Gap
1	Amir Hossein Nabizadeh, José Paulo Leal, Hamed N. Rafsanjani, Rajiv Ratn Shah	2021	Existing e-learning systems fail to provide personalized learning paths for different learners.	Learning Path Personalization, Recommender Systems, Intelligent Tutoring Systems	Lacks real-time learner progress analysis and adaptive recommendations. (ScienceDirect)
2	Yun Yue, Huan Dai, Yupei Zhang, Xuequn Shang, Zhanhuai Li	2022	Traditional learning systems cannot efficiently recommend personalized learning paths.	Learning Analytics, Artificial Intelligence, Adaptive Learning Algorithms	Limited integration of learner behavior with dynamic recommendation models. (JOS)
3	Xianyu Chen, Jian Shen, Wei Xia, et al.	2023	Existing recommendation methods ignore relationships among learning concepts.	Concept-Aware Recommendation, Set-to-Sequence Ranking, Deep Learning	Does not provide complete learner skill analytics and progress visualization. (AAAI Open Access Journal)
4	Feng Zhang, Xuguang Feng, Yibing Wang	2024	Fixed learning paths cannot adapt to learners' changing knowledge levels.	Process Mining, Deep Knowledge Tracing, Personalized Recommendation	High computational complexity and limited scalability for large educational datasets. (ScienceDirect)
5	Yaqian Zheng, Deliang Wang, Junjie Zhang, Yanyan Li, et al.	2025	Existing recommendation systems have limited accuracy in generating personalized learning pathways.	Hybrid Recommendation Framework, Learner Profile Analysis, Knowledge-Based Recommendation	Requires better real-time skill tracking and explainable recommendations. (eric.ed.gov)

Existing vs Proposed system

Existing System	Proposed System
Same content for everyone	Personalized learning path
Manual progress tracking	Automatic skill analytics
No recommendations	AI-based recommendations
Limited reports	Interactive dashboards
Static learning	Adaptive learning
Poor engagement	Improved learning experience

Advantages

Personalized recommendations

Better learner engagement

Easy progress tracking

Novelty

Combines Skill Analytics and Recommendation Engine in one system.



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Requirements



Requirements

Functional Requirements

- User Login
- Registration
- Course Management
- Progress Tracking
- Recommendation Engine
- Reports

Non-Functional Requirements

- Security
- Reliability
- Scalability
- Performance
- Usability

Hardware

- Intel i5/i7 Processor
- 8GB RAM
- 256GB SSD

Software

- Java
- MySQL
- JDBC
- NetBeans / Eclipse
- Windows 10/11

Dataset/APIs

Student learning records

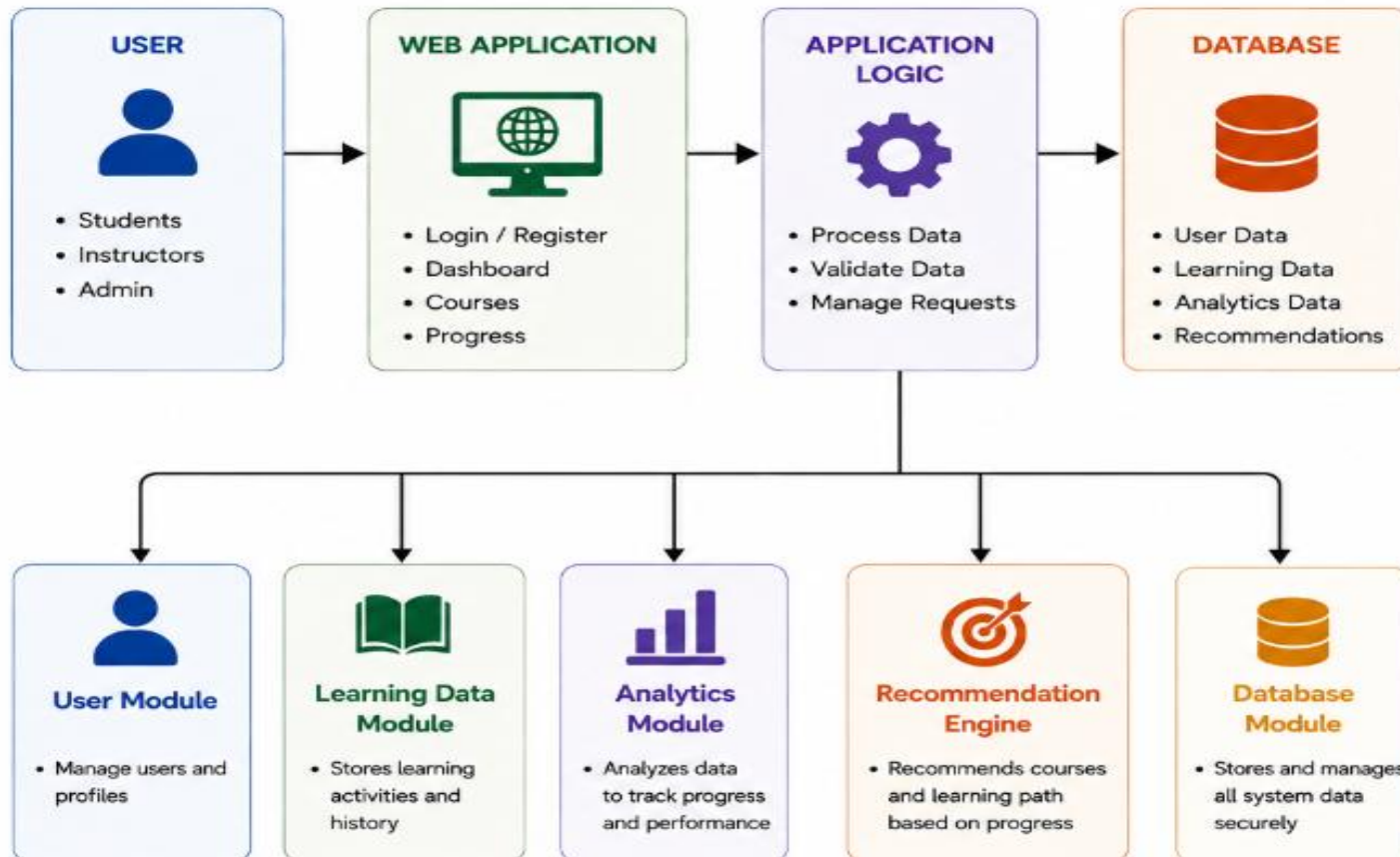
Quiz scores

Course completion data



System Architecture

BLOCK DIAGRAM



WORKFLOW





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Planning



Work Breakdown Structure (WBS) Gantt Chart

- Requirement Analysis
- System Design
- Java Development
- Database Design
- Testing
- Documentation
- Final Deployment

Task	Month 1	Month 2	Month 3	Month 4
Requirement Analysis	✓			
Design		✓		
Development			✓	
Testing				✓
Documentation				✓

Milestones

- Literature Survey Completed
- Design Completed
- Development Completed
- Testing Completed
- Final Presentation

Risk Analysis

- Data quality issues
- Requirement changes
- Time constraints
- Database errors

Expected Outcomes

Deliverables

Java Application

MySQL Database

Progress Dashboard

Project Documentation

Benefits

Personalized learning experience

Better skill improvement

Increased learner engagement

Improved learning efficiency

References (IEEE)

Learning Path Personalization Survey, IEEE, 2021.

Learning Analytics Based Recommendation, IEEE, 2022.

Graph-Based Learning Recommendation, IEEE, 2023.

Personalized Learning Using Deep Knowledge Tracing, IEEE, 2024.

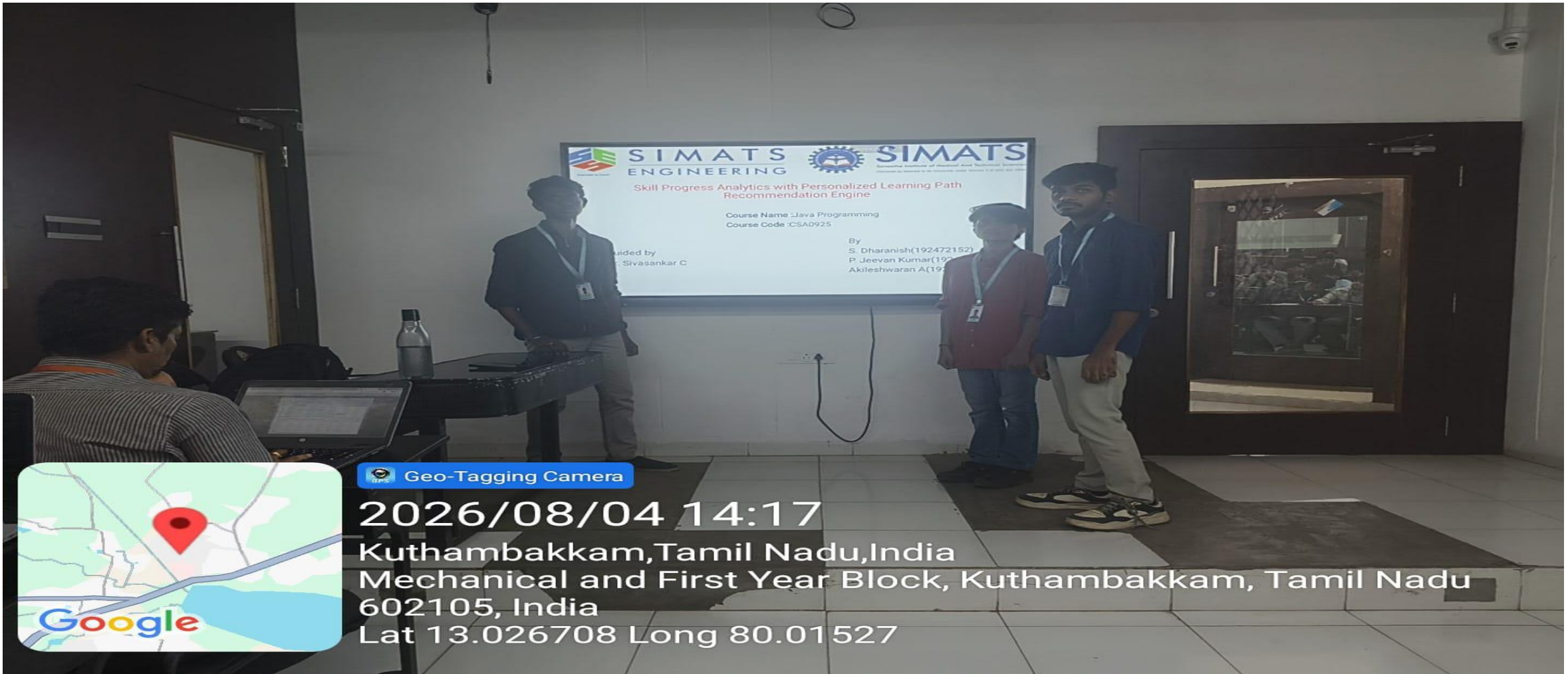
Knowledge Graph Based Personalized Learning, IEEE, 2025.



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Geo Tag Photo



THANK YOU